

Ransomware Incident Response & Recovery

An evidence-led Windows lab case study

From live file monitoring to verified restoration: investigating ransomware-like artifacts with synthetic Finance data and a reversible Base64 simulation.

IMPACT	RECOVERY	INTEGRITY
4 synthetic files	4 files restored	4 / 4 hashes matched

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Defensive portfolio project | Evidence-reviewed edition

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Northstar Ridge is used as the case-study label. This report describes a controlled training exercise, not a production incident or an engagement for a verified organization.

Safety boundary: no real ransomware and no cryptographic encryption.

01 Executive summary

The exercise demonstrated a complete defensive workflow: establish a baseline, observe a controlled file-impact event, preserve investigative records, assess scope, validate backups, restore data, and prove recovery integrity.

A PowerShell simulator changed four synthetic Finance text files, appended **.locked** to their names, and created a recovery-note artifact. Live FileSystemWatcher records and process observations linked the activity to the simulator. SHA-256 comparisons showed that all four file contents changed. [E06, E14, E15]

All four backup copies matched the pre-event baseline. After restoration, all four recovered files matched that same baseline; the completion record reported zero locked files and no recovery note. [E21, E25, E26]

Measure	Evidence-backed result
Affected files / changed hashes	4 / 4
Validated backups / restored hash matches	4 / 4
Residual .locked files / recovery note	0 / absent
Real encryption	None; Base64 simulation

Business relevance

The lab illustrates how an analyst can move from visible file disruption to a defensible recovery decision. The value is the evidence chain and validation method; the exercise does not establish production-scale resilience, business downtime, or financial loss.

Reading guide

Overview and methodology: pages 3-4. Timeline and findings: pages 5-6. Containment and recovery: page 7. ATT&CK, remediation, and skills: pages 8-9. Evidence references: page 10.

02 Incident overview

The target was the dedicated TestData directory in the Windows Ransomware-IR-Lab workspace. The recovery source was the separate PreSimulationBackup directory. Both were local lab folders; this was not a test of offline or immutable backups.

Synthetic Finance file	Observed impact
Budget_Projections.txt	Content changed; .locked appended
Finance_Q3_Forecast.txt	Content changed; .locked appended
Payroll_Reconciliation.txt	Content changed; .locked appended
Vendor_Payments.txt	Content changed; .locked appended

Execution and impact

The script read each text file, converted its content to Base64, wrote a simulation marker and encoded content, renamed the file, and created README_RECOVERY_INSTRUCTIONS.txt. Script review confirms that this transformation was encoding, not encryption. [S01]

Scope assessment

At 12:57:26, the scope export recorded four locked files, a recovery note, and available original backups. This supports the observed TestData scope. It does not constitute a comprehensive search of the host or network. [E19, E20]

Analyst guardrails

Use synthetic data and a dedicated target, validate paths, retain a known-good baseline and backup, and preserve records before cleanup. The reviewed script contains a target-path equality check; this is a basic lab guardrail rather than hardened isolation. No credential theft, persistence, lateral movement, or exfiltration is implemented in the reviewed simulator. [S01]

03 Methodology and evidence quality

Phase	Analyst method	Evidence
Baseline	Record SHA-256 hashes, inventories, host context, processes and connections.	E01-E05
Observe	Monitor filesystem events and capture simulator process details.	E06-E09, E15
Investigate	Compare before/after hashes; retain note content and hash; inventory scope.	E10-E14, E19-E20
Contain	Query for simulator processes; terminate only if still active.	E18, T01
Recover	Validate backups, restore originals, compare recovered hashes.	E21-E26
Report	Reconstruct timeline and document findings and controls.	E27-E33

Independent cross-checks

This report compares the hash values in the baseline, post-event comparison, backup validation, and recovery validation exports rather than relying only on the project summary. Filesystem records support the four renames and recovery-note creation. Process records support the execution mechanism.

Limits and confidence

The records are locally generated CSV exports and a PowerShell transcript, not a forensic disk image or tamper-evident collection. A changed hash proves changed bytes; it does not prove encryption or malicious intent. Process timing and script behavior support attribution within the controlled exercise.

Eight original screenshots are reproduced in the visual evidence appendix. Tables elsewhere reproduce CSV evidence values. The public portfolio website is linked on the cover and closing page.

04 Evidence and incident timeline

Recorded time	Event	Source
12:52:23	PowerShell simulator starts; PID 1732.	E09
12:52:31.735	Budget_Projections.txt content changes.	E06
12:52:32.490	Budget_Projections.txt renamed .locked.	E06
12:52:33.241 / 33.992	Finance_Q3_Forecast.txt changes / renames.	E06
12:52:34.743 / 35.495	Payroll_Reconciliation.txt changes / renames.	E06
12:52:35.258	Process snapshot captures simulator command line.	E15
12:52:36.246 / 36.997	Vendor_Payments.txt changes / renames.	E06
12:52:37.749	Recovery-note artifact created.	E06
12:57:26	Scope recorded: four locked files and note present.	E20
13:00:42	Recovery start and completion recorded.	E22, E26

Timeline interpretation

All timestamps are dated October 7, 2026 and reproduced as local lab times. The exports do not include a timezone offset. Monitor tests at 12:48-12:49 are excluded from incident impact. Recovery start and completion share a one-second timestamp; their logical order follows the transcript, and subsecond recovery duration cannot be established. [E06, T01]

Process evidence

powershell.exe | PID 1732 | parent PID 35996

Arguments: **-NoProfile -ExecutionPolicy Bypass -File**

Script: **simulate_ransomware.ps1** [E15]

05 Findings RANSOM-001 to RANSOM-008

RANSOM-001 | Synthetic Finance files were modified

High | Confirmed. All four post-event SHA-256 hashes differed from their known-good baseline hashes. The controlled simulation altered file contents. [E01, E14]

RANSOM-002 | Synthetic Finance files were rapidly renamed

High | Confirmed. FileSystemWatcher recorded four files renamed with the .locked extension. The behavior reproduced a common ransomware-impact artifact. [E06]

RANSOM-003 | Recovery-note artifact was created

Moderate | Confirmed. README_RECOVERY_INSTRUCTIONS.txt was created and preserved with a SHA-256 hash. The exercise reproduced a ransom-note-style filesystem artifact. [E06, E12, E13]

RANSOM-004 | PowerShell process executed the simulation

High | Confirmed. Process monitoring captured powershell.exe running simulate_ransomware.ps1 with NoProfile and ExecutionPolicy Bypass. PowerShell was the confirmed execution mechanism for the controlled exercise. [E08, E09, E15]

RANSOM-005 | Incident scope was limited to four synthetic files

Informational | Confirmed. Scope assessment identified four .locked files inside the dedicated TestData directory. Four affected files were identified in TestData; the collected scope evidence does not establish host-wide absence of impact. [E19, E20]

RANSOM-006 | Known-good backup copies remained intact

Informational | Confirmed. All four backup SHA-256 hashes matched the pre-incident baseline. A cryptographically verified recovery source was available. [E21]

RANSOM-007 | Recovery successfully restored all affected files

Informational | Confirmed. All restored files matched original baseline SHA-256 hashes after recovery. Recovery integrity was cryptographically validated. [E24, E25, E26]

RANSOM-008 | Real ransomware encryption was not performed

Informational | Guardrail. The simulator used reversible Base64 encoding only inside the dedicated training directory. The project demonstrates ransomware-like forensic behavior without executing destructive malware. [S01]

Severity labels describe simulated scenario significance. They are inherited from the lab findings and do not represent production risk scores.

06 Containment and verified recovery

Containment decision

The transcript records a query for processes whose command line matched the simulator, plus conditional Stop-Process logic. E18 is a zero-byte export. The evidence supports a process check but does not document an executed termination or host isolation. The short-lived script may already have exited. [E18, T01, S01]

Recovery sequence

1. Compare all four backup hashes with the original baseline.
2. Document the affected-file inventory and recovery-note evidence.
3. Remove simulated locked files and the recovery note from TestData.
4. Restore the original Finance files from PreSimulationBackup.
5. Generate restored hashes and compare them with the baseline.
6. Check the final inventory and residual artifact counts. [E19, E21-E26, T01]

File	Changed after event	Backup match	Restored match
Budget_Projections.txt	Yes	Yes	Yes
Finance_Q3_Forecast.txt	Yes	Yes	Yes
Payroll_Reconciliation.txt	Yes	Yes	Yes
Vendor_Payments.txt	Yes	Yes	Yes

Acceptance criteria met

Four restored files; four baseline-matching SHA-256 values; zero remaining .locked files; no remaining recovery note. These checks establish byte-level recovery integrity for the four synthetic files. They do not test application functionality, permissions, production recovery objectives, or eradication of real malware. [E25, E26]

Preservation boundary

Hash records, file inventories, process observations and recovery-note text are available. A separate retained copy of every impacted file was not supplied for this review; do not describe this package as a complete forensic acquisition.

07 Conservative MITRE ATT&CK; mapping

Technique / tactic	Lab status	Evidence and interpretation
T1059.001 PowerShell Execution	Observed	powershell.exe and the simulator command line were captured. This establishes PowerShell execution within the lab.
T1486 Data Encrypted for Impact Impact	Safely emulated	File modification, .locked renaming and a recovery note reproduce impact artifacts. Base64 is not cryptographic encryption; T1486 was not actually executed.

Mapping discipline

Only the observed execution mechanism and deliberately emulated impact pattern are mapped. ExecutionPolicy Bypass alone does not establish defense evasion. Neither Base64 file content nor network snapshots justify additional techniques without supporting behavior. [E15, E31, S01]

Authoritative definitions

MITRE ATT&CK: [T1059.001 - PowerShell](#)

MITRE ATT&CK: [T1486 - Data Encrypted for Impact](#)

Definitions checked October 7, 2026.

Detection concept

Correlate a script interpreter starting with a burst of file writes, repeated extension changes, and note creation in the same scope. Preserve command lines and file events for triage. Treat the correlation as a hypothesis to validate and tune; a legitimate batch job can produce similar signals.

08 Lessons, remediation and skills

Priority	Recommended improvement	How to validate
1	Retain centralized process, command-line and PowerShell logs alongside file events.	Repeat the lab and reconstruct execution and impact from retained logs.
1	Keep protected backups separated from ordinary write credentials; test restores.	Restore a test set and compare its baseline hashes.
2	Apply least privilege and application control appropriate to the environment.	Verify authorized workflows and blocked unauthorized execution.
2	Tune behavioral detections for correlated file impact.	Test both the simulator and legitimate bulk file operations.
2	Strengthen collection with explicit no-match process results, timezone offsets and evidence manifests.	Confirm each collection records success, empty results and file hashes.
3	Exercise production response playbooks and recovery objectives.	Measure approved drills; document functional checks and recovery times.

What the exercise taught

Baselines make change measurable. Process context explains the execution mechanism. Evidence collection must precede cleanup. Backup integrity and recovered-file integrity require separate checks. Precise reporting distinguishes observed facts, emulation, and untested production assumptions. [E32]

Skills demonstrated

Investigation: Windows process analysis, live file monitoring, scope assessment, timeline reconstruction and evidence interpretation.

Response: containment checks, backup validation, restoration and SHA-256 integrity verification.

Communication: executive reporting, traceable findings, conservative ATT&CK mapping and remediation planning.

09 Evidence reference index

References E01-E33 correspond to the numbered filenames in the lab Evidence folder. The report draws on the original local records without modifying them.

IDs	Source files / purpose
E01-E05	Baseline_File_Hashes; Baseline_Processes; Baseline_Network_Connections; Host_Information; Baseline_File_Inventory.
E06-E09	Live_File_Activity; Simulator_Script_Hash; Simulator_Process_Observations; Simulator_Process_Start.
E10-E15	PostSimulation_File_Hashes; PostSimulation_File_Inventory; Recovery_Note_Content; Recovery_Note_Hash; File_Hash_Comparison; Simulator_Process_Summary.
E16-E20	PostSimulation_Processes; PostSimulation_Network_Connections; Containment_Process_Check (empty); Incident_Scope_File_Inventory; Incident_Scope_Summary.
E21-E26	Backup_Validation; Recovery_Start; PostRecovery_File_Inventory; PostRecovery_File_Hashes; Recovery_Validation; Recovery_Completion.
E27-E33	Incident_Timeline; Incident_Findings; Executive_Incident_Summary; Clean_Incident_Timeline; MITRE_ATTACK_Mapping; Lessons_Learned_and_Remediation; Project_Results_Summary.
T01 / S01	PowerShell-Transcript.txt / simulate_ransomware.ps1.

Integrity cross-check

Budget_Projections.txt

Baseline = backup = restored SHA-256:

8A450A556577EB4D6B68CFFBED151037304E463D3A0ABF99841048555A43CC4A

Finance_Q3_Forecast.txt

Baseline = backup = restored SHA-256:

2DCA6EE68BA87757F4E6B40B453F1FD0CCD1960944269FF793F1301748EFF32D

Payroll_Reconciliation.txt

Baseline = backup = restored SHA-256:

4F60520852D68AF26E05189D643F406ECB2B2D3FB0B095BC0ED6D18596CD7C0E

Vendor_Payments.txt

Baseline = backup = restored SHA-256:

136C0C9A8BCD8FFEA784CF5BCB148826D68822AC3E1204C81E10023DF33D6322

Full post-event hash values are retained in E14. All four comparisons were cross-checked for this report.

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Figure 1 - Live monitoring setup

```

PS C:\Users\ryqnb> Write-Host " LIVE FILE ACTIVITY MONITOR"
LIVE FILE ACTIVITY MONITOR
PS C:\Users\ryqnb> Write-Host "=====
=====
PS C:\Users\ryqnb> Write-Host "Watching: $TestDataPath"
Watching: C:\Users\ryqnb\Desktop\Ransomware-IR-Lab\TestData
PS C:\Users\ryqnb> Write-Host "Logging: $MonitorLog"
Logging: C:\Users\ryqnb\Desktop\Ransomware-IR-Lab\Evidence\06_Live_File_Activity.csv
PS C:\Users\ryqnb> Write-Host ""

PS C:\Users\ryqnb> Write-Host "Leave this window running."
Leave this window running.
PS C:\Users\ryqnb> Write-Host "Press Ctrl+C when the incident simulation is finished."
Press Ctrl+C when the incident simulation is finished.
PS C:\Users\ryqnb> Write-Host "=====
=====
PS C:\Users\ryqnb> Write-Host ""

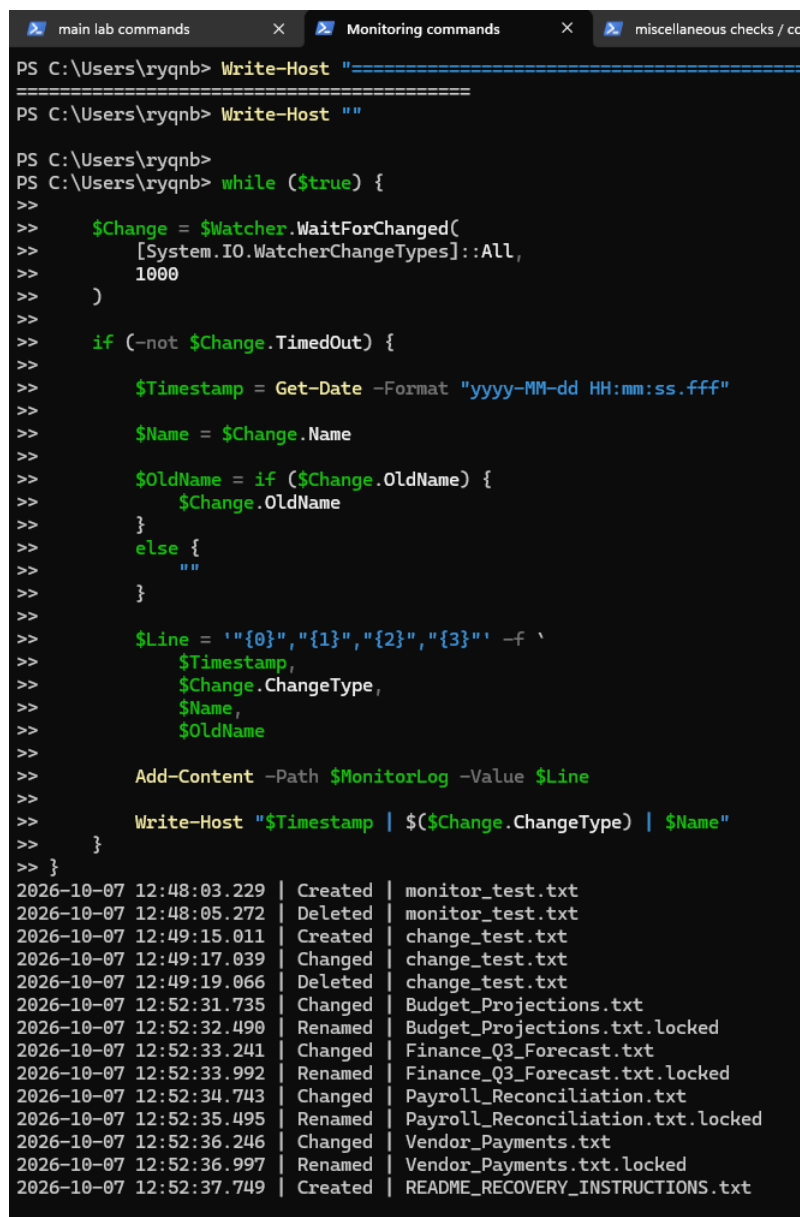
PS C:\Users\ryqnb>
PS C:\Users\ryqnb> while ($true) {
>>
>>     $Change = $Watcher.WaitForChanged(
>>         [System.IO.WatcherChangeTypes]::All,
>>         1000
>>     )
>>
>>     if (-not $Change.TimedOut) {
>>
>>         $Timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss.fff"
>>
>>         $Name = $Change.Name
>>
>>         $OldName = if ($Change.OldName) {
>>             $Change.OldName
>>         }
>>         else {
>>             ""
>>         }
>>
>>         $Line = "{0}","{1}","{2}","{3}" -f `
>>             $Timestamp,
>>             $Change.ChangeType,
>>             $Name,
>>             $OldName
>>
>>         Add-Content -Path $MonitorLog -Value $Line
>>
>>         Write-Host "$Timestamp | $($Change.ChangeType) | $Name"
>>     }
>> }
  
```

The console shows the dedicated TestData watch target, CSV output path, and FileSystemWatcher loop. This documents monitoring configuration; recorded impact appears in Figure 2.

Source: Screenshot 2026-10-07 124712.png. Evidence cross-reference: E06, T01. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

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Figure 2 - Observed file activity



```

main lab commands  Monitoring commands  miscellaneous checks / co
PS C:\Users\ryqnb> Write-Host "=====
=====
PS C:\Users\ryqnb> Write-Host ""

PS C:\Users\ryqnb>
PS C:\Users\ryqnb> while ($true) {
>>     $Change = $Watcher.WaitForChanged(
>>         [System.IO.WatcherChangeTypes]::All,
>>         1000
>>     )
>>     if (-not $Change.TimedOut) {
>>         $Timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss.fff"
>>         $Name = $Change.Name
>>         $OldName = if ($Change.OldName) {
>>             $Change.OldName
>>         }
>>         else {
>>             ""
>>         }
>>         $Line = "{0}|{1}|{2}|{3}" -f `
>>             $Timestamp,
>>             $Change.ChangeType,
>>             $Name,
>>             $OldName
>>         Add-Content -Path $MonitorLog -Value $Line
>>         Write-Host "$Timestamp | $($Change.ChangeType) | $Name"
>>     }
>> }
2026-10-07 12:48:03.229 | Created | monitor_test.txt
2026-10-07 12:48:05.272 | Deleted | monitor_test.txt
2026-10-07 12:49:15.011 | Created | change_test.txt
2026-10-07 12:49:17.039 | Changed | change_test.txt
2026-10-07 12:49:19.066 | Deleted | change_test.txt
2026-10-07 12:52:31.735 | Changed | Budget_Projections.txt
2026-10-07 12:52:32.490 | Renamed | Budget_Projections.txt.locked
2026-10-07 12:52:33.241 | Changed | Finance_Q3_Forecast.txt
2026-10-07 12:52:33.992 | Renamed | Finance_Q3_Forecast.txt.locked
2026-10-07 12:52:34.743 | Changed | Payroll_Reconciliation.txt
2026-10-07 12:52:35.495 | Renamed | Payroll_Reconciliation.txt.locked
2026-10-07 12:52:36.246 | Changed | Vendor_Payments.txt
2026-10-07 12:52:36.997 | Renamed | Vendor_Payments.txt.locked
2026-10-07 12:52:37.749 | Created | README_RECOVERY_INSTRUCTIONS.txt

```

The console records content changes and .locked renames for all four synthetic Finance files, followed by recovery-note creation. Earlier monitor_test and change_test events are setup tests, not incident impact.

Source: Screenshot 2026-10-07 125346.png. Evidence cross-reference: E06. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

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Figure 3 - Process collection workflow

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\ryqnb> $LabPath = "$env:USERPROFILE\Desktop\Ransomware-IR-Lab"
PS C:\Users\ryqnb> $EvidencePath = "$LabPath\Evidence"
PS C:\Users\ryqnb>
PS C:\Users\ryqnb> Write-Host "Incident monitoring console ready."
Incident monitoring console ready.
PS C:\Users\ryqnb> $LabPath = "$env:USERPROFILE\Desktop\Ransomware-IR-Lab"
PS C:\Users\ryqnb> $EvidencePath = "$LabPath\Evidence"
PS C:\Users\ryqnb> $ProcessLog = "$EvidencePath\08_Simulator_Process_Observations.csv"
PS C:\Users\ryqnb>
PS C:\Users\ryqnb> Write-Host "Watching for simulate_ransomware.ps1..."
Watching for simulate_ransomware.ps1...
PS C:\Users\ryqnb> Write-Host "Switch to Tab 1 and launch the simulator now."
Switch to Tab 1 and launch the simulator now.
PS C:\Users\ryqnb>
PS C:\Users\ryqnb> $Observations = for ($i = 0; $i -lt 45; $i++) {
>>
>>     Get-CimInstance Win32_Process |
>>         Where-Object {
>>             $_.CommandLine -match "simulate_ransomware\.ps1"
>>         } |
>>         ForEach-Object {
>>
>>             [PSCustomObject]@{
>>                 Timestamp      = Get-Date -Format "yyyy-MM-dd HH:mm:ss.fff"
>>                 ProcessId       = $_.ProcessId
>>                 ParentProcessId = $_.ParentProcessId
>>                 ProcessName     = $_.Name
>>                 CommandLine     = $_.CommandLine
>>             }
>>         }
>>
>>     Start-Sleep -Seconds 1
>> }
PS C:\Users\ryqnb>
PS C:\Users\ryqnb> $Observations |
>>     Export-Csv $ProcessLog -NoTypeInformation
PS C:\Users\ryqnb>
PS C:\Users\ryqnb> Write-Host "Process monitoring complete."
Process monitoring complete.
PS C:\Users\ryqnb> Write-Host "Saved to: $ProcessLog"

```

The console shows Win32_Process polling for the simulator command line and export to the process-observation CSV. This screenshot documents the collection method; PID and command-line results appear in Figure 4.

Source: Screenshot 2026-10-07 125401.png. Evidence cross-reference: E08, E15. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

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Figure 4 - Post-event hashes and process evidence

```

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Write-Host " POST-INCIDENT EVIDENCE COLLECTION COMPLETE"
POST-INCIDENT EVIDENCE COLLECTION COMPLETE
PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Write-Host "=====
=====
PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Write-Host ""

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Get-ChildItem $EvidencePath |
>> Sort-Object Name |
>> Select-Object Name, Length

Name                                     Length
----
01_Baseline_File_Hashes.csv             637
02_Baseline_Processes.csv               16065
03_Baseline_Network_Connections.csv     5643
04_Host_Information.txt                  386
05_Baseline_File_Inventory.csv           366
06_Live_File_Activity.csv                1044
07_Simulator_Script_Hash.csv             171
08_Simulator_Process_Observations.csv    4283
09_Simulator_Process_Start.csv           77
10_PostSimulation_File_Hashes.csv        827
11_PostSimulation_File_Inventory.csv      488
12_Recovery_Note_Content.txt             468
13_Recovery_Note_Hash.csv                189
14_File_Hash_Comparison.csv              871
15_Simulator_Process_Summary.csv         385
16_PostSimulation_Processes.csv          16162
17_PostSimulation_Network_Connections.csv 6176

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Import-Csv "$EvidencePath\14_File_Hash_Comparison.csv" |
>> Format-Table -AutoSize

OriginalFile                           ModifiedFile                           BaselineHash                           PostHash                               HashChanged
-----
Budget_Projections.txt                 Budget_Projections.txt.locked           8A458A556577EB4D6868CFFBED151037304E463D3A0ABF99841048555A43CC4A 29D5FD89708C22B955C7E934465212DCF23D488C06EEA0B2CAA00FB9E391D655 True
Finance_Q3_Forecast.txt                Finance_Q3_Forecast.txt.locked           2DCA6EE68BA37757F4E6800B43F1F0CCD1060944269F793F1301748EFF32D F0A328FE4441D85BDEE2A2C013F80B343644C4BFDB3746E87032CDE9266E908 True
Payroll_Reconciliation.txt              Payroll_Reconciliation.txt.locked        4F60520852D68AF26E051B90643F406ECE8282D3F8080995DC0ED4018596CD7C0E 85F7A571665906123FC7C80999E4F8E217E4F3008FEC44D9E1D37C13A118158F True
Vendor_Payments.txt                   Vendor_Payments.txt.locked              136C0C9A8BCD8FFEA784CF58CB148826D68822AC3E1204C81E100230F33D6322 F28F07F732544534CA68EF706C745769F09E8F869266F86C58FC809F2F6C6AB True

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Import-Csv "$EvidencePath\15_Simulator_Process_Summary.csv" |
>> Format-Table -AutoSize

Timestamp                               ProcessId ParentProcessId ProcessName CommandLine
-----
2026-10-07 12:52:35.258 1732          35996 powershell.exe "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -ExecutionPolicy Bypass -File "C:\Users\ryqnb\Desktop\Ransomware-IR-L...
PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab>

```

The table shows four changed hashes and powershell.exe PID 1732 with parent PID 35996. The displayed command line is truncated; E15 retains the complete command line.

Source: Screenshot 2026-10-07 125626.png. Evidence cross-reference: E14, E15. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

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Figure 5 - Scope and backup validation

```
Timestamp          : 2026-10-07 12:57:26
AffectedDirectory  : C:\Users\ryqnb\Desktop\Ransomware-IR-Lab\TestData
LockedFileCount    : 4
RecoveryNotePresent : True
OriginalBackupPresent : True

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Import-Csv "$EvidencePath\21_Backup_Validation.csv" |
>> Format-Table -AutoSize

FileName           BaselineHash                                     BackupHash                                     BackupHashMatches
-----
Budget_Projections.txt 8A450A556577EB4D6B68CFFBBD151037304E463D3A0ABF99841048555A43CC4A 8A450A556577EB4D6B68CFFBBD151037304E463D3A0ABF99841048555A43CC4A True
Finance_Q3_Forecast.txt 2DCA6EE68BA87757F4E6B40B453F1FD0CCD1960944269FF793F1301748EFF32D 2DCA6EE68BA87757F4E6B40B453F1FD0CCD1960944269FF793F1301748EFF32D True
Payroll_Reconciliation.txt 4F60520852D68AF26E05189D643F406ECB2B2D3FB080958C0ED6D18596CD7C0E 4F60520852D68AF26E05189D643F406ECB2B2D3FB080958C0ED6D18596CD7C0E True
Vendor_Payments.txt 136C0C9A8BCD8FFEA784CF5BCB148826D68822AC3E1204C81E100230F33D6322 136C0C9A8BCD8FFEA784CF5BCB148826D68822AC3E1204C81E100230F33D6322 True

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> |
```

Scope output records four locked files and a recovery note in TestData. All four backup hashes match their baseline values before restoration.

Source: Screenshot 2026-10-07 125936.png. Evidence cross-reference: E20, E21. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

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Figure 6 - Recovery validation and restored inventory

FileName	BaselineHash	RestoredHash	RecoveryValidated
Budget_Projections.txt	8A450A556577EB4D6B68CFFBED151037304E463D3A0ABF99841048555A43CC4A	8A450A556577EB4D6B68CFFBED151037304E463D3A0ABF99841048555A43CC4A	True
Finance_Q3_Forecast.txt	2DCA6EE68BA87757F4E6840B453F1FD0CCD1960944269FF793F1301748EFF32D	2DCA6EE68BA87757F4E6840B453F1FD0CCD1960944269FF793F1301748EFF32D	True
Payroll_Reconciliation.txt	4F60520852D68AF26E05189D643F406ECB282D3FB0B095BC0ED6D18596CD7C0E	4F60520852D68AF26E05189D643F406ECB282D3FB0B095BC0ED6D18596CD7C0E	True
Vendor_Payments.txt	136C0C9A8BCD8FFEA784CF5BCB148826D68822AC3E1204C81E10023DF33D6322	136C0C9A8BCD8FFEA784CF5BCB148826D68822AC3E1204C81E10023DF33D6322	True


```

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Import-Csv "$EvidencePath\26_Recovery_Completion.csv" |
>> Format-List

RecoveryEnd      : 10/7/2026 1:00:42 PM
LockedFilesRemaining : 0
RecoveryNoteRemaining : False
RestoredFileCount  : 4

PS C:\Users\ryqnb\Desktop\Ransomware-IR-Lab> Get-ChildItem $TestDataPath |
>> Select-Object Name, Length

```

Name	Length
Budget_Projections.txt	56
Finance_Q3_Forecast.txt	63
Payroll_Reconciliation.txt	55
Vendor_Payments.txt	51

All four recovery comparisons are True. Completion output records zero locked files, no recovery note, and four restored files; the final inventory lists the original filenames.

Source: Screenshot 2026-10-07 130428.png. Evidence cross-reference: E23, E25, E26. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

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Figure 7 - Original incident timeline

```
>> Format-Table Timestamp, Phase, Event -AutoSize
```

Timestamp	Phase	Event
2026-10-07 12:48:03.229	Impact	Created - monitor_test.txt
2026-10-07 12:48:05.272	Impact	Deleted - monitor_test.txt
2026-10-07 12:49:15.011	Impact	Created - change_test.txt
2026-10-07 12:49:17.039	Impact	Changed - change_test.txt
2026-10-07 12:49:19.066	Impact	Deleted - change_test.txt
10/7/2026 12:52:23 PM	Execution	PowerShell ransomware-behavior simulator started
2026-10-07 12:52:31.735	Impact	Changed - Budget_Projections.txt
2026-10-07 12:52:32.490	Impact	Renamed - Budget_Projections.txt.locked
2026-10-07 12:52:33.241	Impact	Changed - Finance_Q3_Forecast.txt
2026-10-07 12:52:33.992	Impact	Renamed - Finance_Q3_Forecast.txt.locked
2026-10-07 12:52:34.743	Impact	Changed - Payroll_Reconciliation.txt
2026-10-07 12:52:35.495	Impact	Renamed - Payroll_Reconciliation.txt.locked
2026-10-07 12:52:36.246	Impact	Changed - Vendor_Payments.txt
2026-10-07 12:52:36.997	Impact	Renamed - Vendor_Payments.txt.locked
2026-10-07 12:52:37.749	Impact	Created - README_RECOVERY_INSTRUCTIONS.txt
2026-10-07 12:57:26	Containment / Scope	Incident scope confirmed
10/7/2026 1:00:42 PM	Recovery	Recovery completed
10/7/2026 1:00:42 PM	Recovery	Known-good recovery process started

The original display includes monitor setup tests and lists completion before recovery start at the same timestamp. The report timeline excludes setup tests and uses transcript workflow order for the tied recovery records.

Source: Screenshot 2026-10-07 130445.png. Evidence cross-reference: E27, E30, T01. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

17 Visual evidence | 8

Figure 8 - Finding register

```
>> Format-Table FindingID, Severity, Status, Finding -AutoSize
```

FindingID	Severity	Status	Finding
RANSOM-001	High	Confirmed	Synthetic Finance files were modified
RANSOM-002	High	Confirmed	Synthetic Finance files were rapidly renamed
RANSOM-003	Moderate	Confirmed	Recovery-note artifact was created
RANSOM-004	High	Confirmed	PowerShell process executed the simulation
RANSOM-005	Informational	Confirmed	Incident scope was limited to four synthetic files
RANSOM-006	Informational	Confirmed	Known-good backup copies remained intact
RANSOM-007	Informational	Confirmed	Recovery successfully restored all affected files
RANSOM-008	Informational	Guardrail	Real ransomware encryption was not performed

The screenshot shows RANSOM-001 through RANSOM-008 with their lab severity and status labels. The report findings preserve these identifiers while explaining scope and evidence limitations.

Source: Screenshot 2026-10-07 130459.png. Evidence cross-reference: E28. Original screenshot reproduced with aspect ratio preserved; zoom the PDF for dense console output.

18 Interview talking points

A concise project introduction

"I completed a defensive incident-response lab using four synthetic Finance files. A controlled PowerShell script used Base64 encoding and file renaming to generate ransomware-like artifacts. I correlated file activity with process evidence, assessed the observed scope, and verified both the backup source and restored files against the original SHA-256 baseline."

What did you establish?

All four file hashes changed during the simulation. Four backups matched the baseline, and all four restored files matched it after recovery. Final records showed zero locked files and no recovery note.

How did you connect execution to impact?

I used timestamped filesystem events, the captured PowerShell command line and PID, and script review. Together they support the controlled simulator as the execution mechanism.

How did you validate recovery?

I checked the recovery source before restoration and generated fresh hashes afterward. A restored filename alone was not enough; the restored bytes had to match the baseline.

What limits did you acknowledge?

The scope export covered the lab directory, not a comprehensive host-wide investigation. The empty containment export did not prove process termination. Recovery timestamps had only one-second precision.

How did you map ATT&CK;?

T1059.001 was observed through PowerShell process evidence. T1486 was safely emulated: Base64 encoding is not encryption, so I did not claim actual encrypted-for-impact behavior.

What would you improve next?

I would centralize logs, explicitly record no-match process checks, retain timezone offsets and evidence manifests, test protected backups, and add functional recovery checks.

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